
API & CPU tokenizer (BPE)
T4 host mean: 0.206 ms (n=20)
Station 1 (Q_1): admission
Batch / iteration scheduler
Staging queue, forms batch $b \le B$
Station 2 (Q_2): FCFS batcher
Parallel GPU execution streams
CUDA prefill + decode ($\max_{j\le b} L_j$ steps)
Station 3 (Q_3): GPU workers
HTTP req.
Bernoulli λ
S_1
token IDs
$\lambda_1=\lambda$
timer τ
batch $b \le B$
1
c
stream
done
Single-server $Geo/D/1$, S_1=2
Wait $W_{q,1}$, util. $\rho_1 = 2\lambda < 1$
Releases when q_2=B or wait $\ge \tau$
Wait $W_{q,2}$, mean fill $\bar{b} \in [1, B]$
c parallel slots ($Geo/Geo/c$ or $S_3(b)$)
Wait $W_{q,3}$, util. $\rho_3 = \lambda E[S_3]/c < 1$
Delay along a request's path (schematic, not to scale)
$S_3(b)$
$W_{q,1}$
S_1
$W_{q,2}$
$W_{q,3}$
prefill
decode
TTFT $= W_{q,1} + S_1 + W_{q,2} + W_{q,3} + S_{\text{prefill}}(b)$
ITL $= S_{\text{decode}}(b, L_{\max})/L_{\text{gen}} \propto (\max_{j\le b} L_j/L_{\text{gen}}) \cdot t_{\text{step}}(b)$
$W_{\text{soj}} = (W_{q,1} + W_{q,2} + W_{q,3}) + S_1 + S_3(b)$